INCH-POUND
MIL-PRF-3098/17H
w/AMENDMENT 2
4 June 2025
SUPERSEDING
MIL-PRF-3098/17H
w/AMENDMENT 1
9 March 2020

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR37/U

INACTIVE FOR NEW
DESIGN AFTER 4 JUNE 2025

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 90 kHz to 250 kHz; fundamental; noncontrolled; anti-resonance.

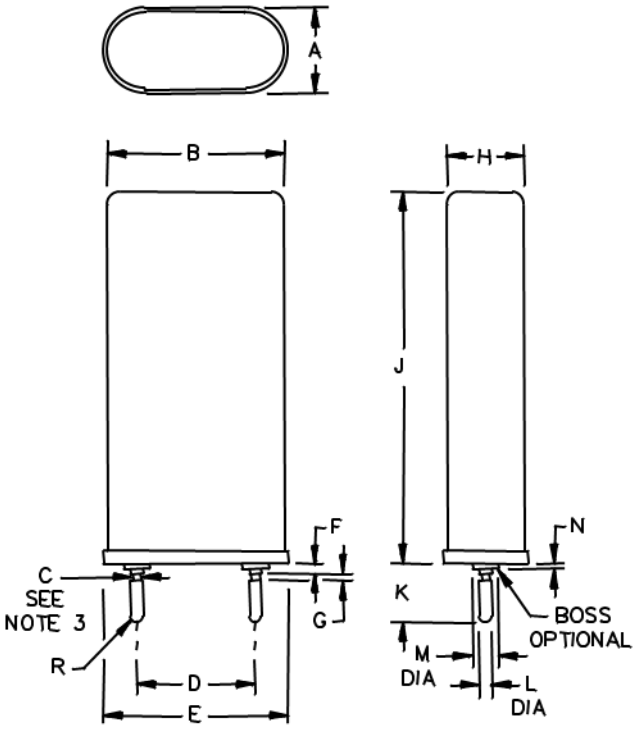


FIGURE 1. Crystal unit - CR37/U.

AMSC N/A
FSC 5955

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.352	---	8.94
B	---	.725	---	18.41
C	.030	.037	0.76	0.94
D	.478	.494	12.14	12.55
E	---	.757	---	19.23
F	.030	.040	0.76	1.02
G	.015	.025	0.38	0.63
K	.223	.248	5.66	6.30
H	---	.317	---	8.05
J	---	1.526	---	38.76
L	.048	.052	1.22	1.32
M	.075	.141	1.90	3.58
N	.015	.025	0.38	0.63

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR37/U - Continued.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 90 kHz to 250 kHz, inclusive.

Capacitance, shunt: See table I.

Frequency tolerance:

Operating temperature range: ±200 parts per million (ppm).

Equivalent resistance: 90 kHz to 170 kHz: 5,000 ohms, maximum.
170+ kHz to 250 kHz: 5,500 ohms, maximum.

Mode of oscillation: Fundamental.

Antiresonance, load capacitance: 20.0 pF ±0.5 pF.

Operating temperature range (noncontrolled): -40°C to +70°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Shock (specified pulse):

Frequency change permitted: ±10 ppm.

Equivalent resistance change permitted: ±15 percent.

Vibration: MIL-STD-202-201, amplitude 0.015 inch (total excursion 0.030 inch).

Frequency change permitted: ±10 ppm.

Equivalent resistance change permitted: ±15 percent.

Temperature run:

Frequency change permitted: ±10 ppm.

Equivalent resistance change permitted: ±15 percent.

Bond strength: 90 kHz to 170 kHz: 800 grams, minimum
170+ kHz to 250 kHz: 700 grams, minimum

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TABLE I. Capacitance, shunt.

Frequency range,	Capacitance 1/
kHz	pF
90 to 170	$\frac{450}{f} + 1.2$
170+ to 250	$\frac{322}{f} + 1.2$

1/ The letter "f" represents specified frequency in kHz. A capacitance tolerance of ±15 percent is permitted.

Part or Identifying Number (PIN): CR37/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Qualification inspection is not applicable for this specification sheet. Acceptance testing shall consist of screening, Group A and Group B inspection of MIL-PRF-3098.

Referenced documents. In addition to MIL-PRF-3098, this document references the following: MIL-STD-202-201.

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy – EC
Air force – 85
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC
(Project 5955-2025-003)

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